

AI Engineer Candidate Assessment: Email Generation Assistant

1. Project Goal: Build an Email Generation Assistant

Objective: Create a working prototype that generates professional emails based on user input, using an LLM.

A. The Assistant's Function

The assistant must take three distinct inputs and produce a well-written, professional email:

1. **Intent:** The core purpose of the email (e.g., "Follow up after meeting," "Request for proposal details.")
2. **Key Facts:** A few bullet points of information that **must** be seamlessly included in the final email.
3. **Tone:** The desired style (e.g., *formal*, *casual*, *urgent*, *empathetic*).

B. Core Technical Requirement: Advanced Prompt Engineering

You must use and document an **advanced prompting technique** (e.g., Role-Playing, Few-Shot Examples, Chain-of-Thought) to maximize output quality and reliability.

2. Evaluation Strategy with Custom Metrics

Objective: Design and implement a system to measure the assistant's performance using metrics you create specifically for this task.

A. Test Data

- Create **10 unique input scenarios** (Intent, Facts, Tone) for the model to generate emails from.
- For each scenario, write a **Human Reference Email** (the ideal output).

B. Custom Evaluation Metrics Implementation

You must define and implement **three (3) custom metrics** designed to judge the quality of an email generation assistant. These should use a combination of automated techniques (e.g., Python scripting, LLM-as-a-Judge) but must be tailored to this specific project.

The Candidate Must Define and Implement:

1. **Custom Metric 1:** _____

- *Example Focus:* Fact Recall, Specificity, or Clarity.
- 2. **Custom Metric 2:** _____
 - *Example Focus:* Tone Accuracy, Format Adherence, or Sentence Structure Complexity.
- 3. **Custom Metric 3:** _____
 - *Example Focus:* Conciseness, Introduction Effectiveness, or Grammar/Fluency.

C. Evaluation Report

Your script must run the 10 scenarios and output a structured file (CSV, JSON, etc.) that clearly shows:

- The **Definition and Logic** for each of your 3 Custom Metrics.
- The raw scores for **all 10 scenarios** for **all 3 Custom Metrics**.
- The overall **average score** for the model based on your metrics.

3. Model Comparison and Analysis

Objective: Run your evaluation against a second model/strategy and analyze the results.

1. **Run Two Models:** Use your **exact same 10 scenarios** and **Evaluation Strategy** (using your 3 custom metrics) to test a **second LLM or prompting approach** (e.g., Model A vs. Model B).
2. **Analysis:** Write a brief, single-page summary answering:
 - Which model/strategy performed better according to your 3 custom metrics?
 - What was the biggest failure mode of the lower-performing model based on your data?
 - Which model do you recommend for production and why (use your custom metric data to justify it)?

Deliverables

1. **Code Repository:** A link to a GitHub repository with all code and a **README** explaining how to set up and execute the project.
2. **Final Report (PDF/Google Doc):**
 - The **Prompt Template** used.
 - The **Definitions and Logic** for the **3 Custom Metrics**.
 - The raw **Evaluation Data** (your CSV/JSON output).
 - The **Comparative Analysis** summary (Section 3).

